

Novel high voltage spinels for Li-ion batteries cathode materials

Corrado Locati
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*My work is dedicated to all the people who believed in me. . .
... It is dedicated to VASCO ROSSI too, the soundtrack of my life (I'm
part of that generazione di sconvolti che non ha più santi nè eroiiii. . . e
siamo solo noiiiiii)¹*

¹This report is written in L^AT_EX

Contents

1	Introduction	1
1.1	General	1
1.2	Why lithium?	2
1.3	Nano and high voltage spinels	3
2	Experimental	7
2.1	Synthesis	7
2.1.1	Solid state synthesis	7
2.1.2	Modified sol-gel	8
2.1.3	Xerogel	8
2.2	Coin-cell fabrication	9
2.3	Characterization techniques	9
2.3.1	Crystallographic structure	9
2.3.2	Crystallinity and morphology	10
2.3.3	Battery testing	10
3	Results	12
3.1	$\text{LiMg}_{0.1}\text{Ni}_{0.4}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.1$	12
3.1.1	Solid State Synthesis	12
3.1.2	Sol-gel Synthesis	18
3.2	$\text{LiMg}_{0.07}\text{Ni}_{0.43}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.07$	21
3.2.1	Solid State Synthesis	21
3.3	$\text{LiMg}_{0.05}\text{Ni}_{0.45}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.05$	25
3.3.1	Solid State Synthesis	25
3.3.2	Sol-gel Synthesis	26
3.3.3	Xerogel Synthesis	30
4	Summary and Conclusions	36
	Acknowledgments	39

Bibliography	40
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Chapter 1

Introduction

1.1 General

The rechargeable batteries are an important component of a multitude of portable electronic devices, but the battery state of the art is slow-growing in respect to the electronic technology growth. Nevertheless the idea of lithium ion (Li-ion) batteries was so revolutionary that they, together with the Ni-MeH batteries, are completely supplanting the old Ni-Cd rechargeable batteries [1, 2, 3]. This is due moreover for their light weight and high energy density properties.

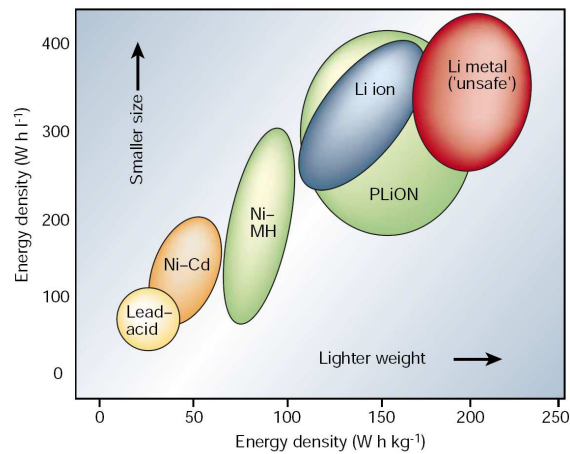


Figure 1.1: Comparison of the different battery technologies in terms of their volumetric and gravimetric energy density.

The various kind of batteries are showed in fig.1.1, in relation with their properties of lightness and energy storage.

A Li-ion battery is made of various electrochemical cells connected in series and/or in parallel, to provide the required voltage or capacity respectively. Each cell consists of a positive and a negative electrode separated by an electrolyte, for example a dissociated salt, in which ions pass from an electrode to the other.

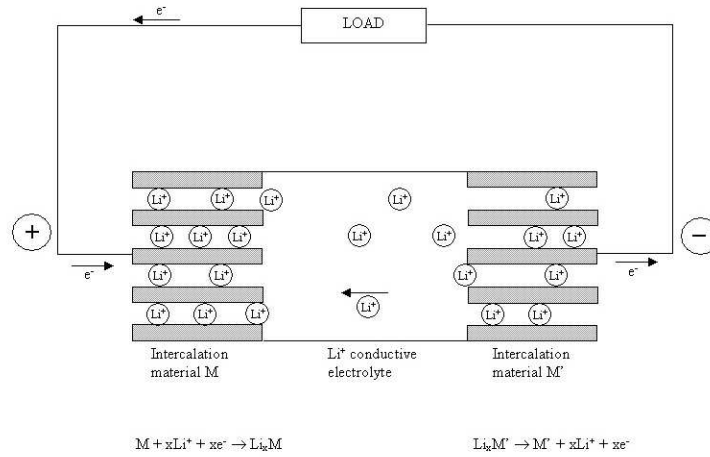


Figure 1.2: Schematic plot of a Li-ion cell.

During the discharge, the electrons go externally from the negative electrode to the positive one, parallel to the lithium ions that go through the electrolyte. During the charge process opposite reactions take place (see fig. 1.2). In this report we will use the terms anode and cathode for the negative and positive electrodes, even though we realize that it is wrongly used. In a rechargeable battery (or secondary cell) the negative electrode is the anode, i.e. the source of electrons, only during the discharge process, but it becomes the cathode during the charge. The same is for the positive electrode: only in the discharge process it acts like a cathode, but during the charge it's an anode.

1.2 Why lithium?

Lithium has a lot of advantages in respect to the other materials, It owns the preferred electrochemical properties: its electrode potential is -3.01 V vs.

SHE (Standard Hydrogen Electrode), which is the lowest value among the metals. It has even the lowest density (0.54 g/cm^3) and the lowest electrochemical equivalent (0.259 gA/h) of all the solids¹. Thanks to these values, batteries with lithium offer the possibility of both high voltage and high energy density.

Lithium in batteries is used both as element and as ion. Primary lithium batteries (primary means *non* rechargeable) uses lithium metal, with organic or stable inorganic lithium solvents selected as electrolyte; lithium manganese oxide, lithium carbon monofluoride or lithium thionyl chloride as cathode materials [4, 5].

In general, lithium-ion batteries have a Li-containing transition metal oxide as positive electrode and a carbon-based material (e.g. graphite) as negative electrode. The cell is assembled in the discharge mode. During the charge, lithium ions pass from the positive electrode through the electrolyte, to the negative electrode. Here they recombines with electrons from the electrical net and the potential of the negative electrode decreases, while the one of the positive electrode rises. That's why electrical energy is stored. In the discharge process, lithium stored in the negative electrode gives up its valence electron and crosses the electrolyte as an ion until the positive electrode is reached. The potential of the negative electrode starts to increase, while the potential of the positive electrode decreases, until their difference reaches the zero or it's limited by an externally set voltage. The battery has done one cycle (of charge-discharge), which can be repeated again. For this reason it is called a *rocking-chair battery*.

1.3 Nano and high voltage spinels

One of the main wishful thinking of the research about Li-ion batteries is to reach the specific power and the specific energy values of a combustion engine. As you can see in the Ragone plot² in fig. 1.3, batteries have a low specific power but a high specific energy [6]. The main task in the Li-ion battery research is to increase them both in power and in capacity. Typically more power means less capacity. In order to improve the power it can be use small (nano) particles because the lithium ions can enter into the nanoparticles

¹The *electrochemical equivalent* is the weight amount of a substance that generates a 1C current, when it reacts over an electrode. Lithium has the lowest value of the electrochemical equivalent and it means that, for a fixed amount of substance, it produces the maximum current in respect all the other substances.

²A Ragone plot shows in a logarithmic scale the specific power i.e. the *capacity* of energy storage versus the specific energy, that is the energy that a device can produce.

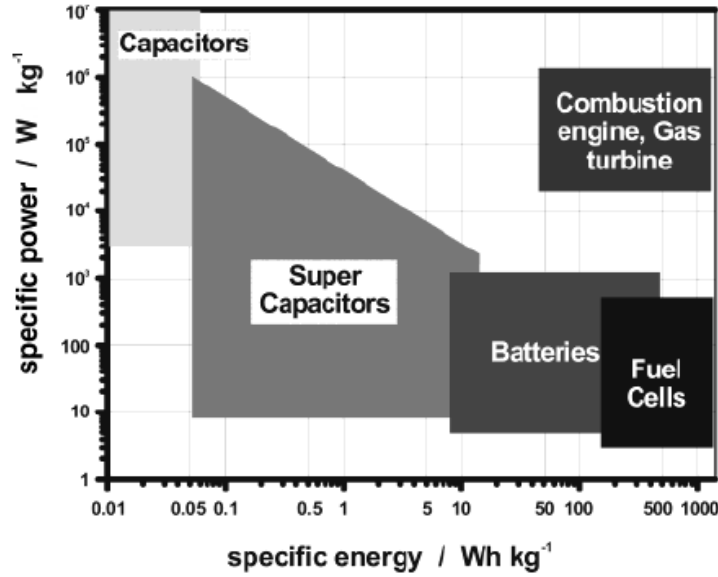
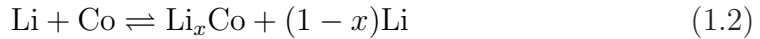


Figure 1.3: Ragone plot of energy storage domains for various electrochemical energy conversion systems devices in comparison with internal combustion engine, turbines and common capacitors.

in less time than a bulk material, thanks to their higher surface area. The nanoparticles have more sites where ions are able to enter in the lattice. Once inside, the ions can spend less time to fill the domains: smaller is the particle and less site for the ions are present, it means that a battery can be charged in less time than now. To increase capacity we need higher voltage with the same capacity in terms of Ah/kg. For anode materials it has been observed that with smaller particles (< 100 nm), the capacity increases due to non-conventional electrochemistry. The successive equations, for example, are reversible only if the products are nano-size:



To increase the voltage we have chosen to operate with spinel structure, like LiMn₂O₄. Such material has less capacity (120 mAh/g) in respect, for example, LiCoO (130 mAh/g). Nevertheless a spinel structure has three dimensional vacancy pathways which permit a better crystal relaxation during

the intercalation process. the reach of high voltage values is hampered by stability problem, both of electrolyte and electrodes. In order to stabilize the electrode, two way are useful: the one is the coating of the powders, with a material that can prevent side reactions on the electrode surface, and the other one is doping the material. We choose the second way, so we started with $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ doped with magnesium. In this work we focused about new high-voltage nanomaterials for positive electrodes, using the formulas $\text{LiMg}_\delta\text{Ni}_{0.5-\delta}\text{Mn}_{1.5}\text{O}_4$ with three different stoichiometries ($\delta=0.05, 0.07, 0.1$). These materials have a cubic spinel structure ($P4_332$) with the following site occupation: Li^+ on 8c, Mg^{2+} on 4b, Ni^{2+} on 4b/12b, Mn^{4+} on 12d/4b and O^{2-} on 24e and 8c (see 1.4) [9].

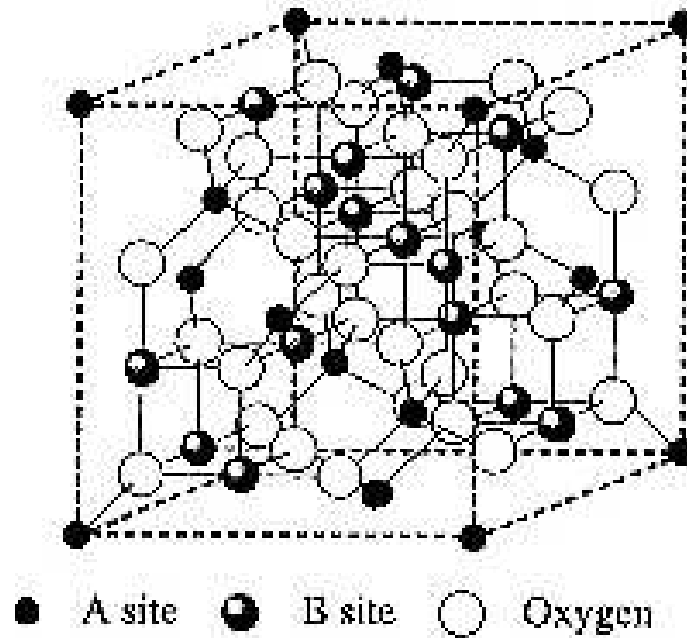
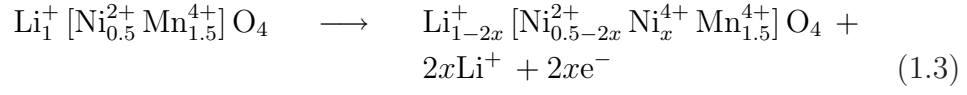


Figure 1.4: Typical spinel structure AB_2O_4 , in which $\text{A} = \text{Li}$; $\text{B} = \text{Mg, Ni, Mn}$; O is the oxygen.

We tried to synthesize and characterize three cathode materials with different amount of magnesium to replace part of the nickel in the spinel. The process of deintercalation (charge of the battery) is, in the case of $\text{Li}_1^+ [\text{Ni}_{0.5}^{2+} \text{Mn}_{1.5}^{4+}] \text{O}_4$: [13, 14]



According the eq. 1.3, only the oxidation of Ni ($\text{Ni}_{0.5}^{2+} \rightarrow \text{Ni}_{0.5-x}^{2+} + \text{Ni}_x^{4+}$) is responsible of deintercalation of lithium. So the discharge capacity would decrease with δ , because the amount of Ni^{2+} decreases and the potential vs. Li/Li^+ would also increase. Mg^{2+} isn't involved on the red-ox process, but it improves the overall conductivity as well as the rate capability and the cycling stability of the material [9].

Chapter 2

Experimental

2.1 Synthesis

The research of these new materials leads us to look for suitable synthesis routes. The cathodes were finally created with three different techniques: a solid state processing, a modified sol-gel synthesis and a xerogel way. The methods have been described as follow.

2.1.1 Solid state synthesis

The first procedure we adopted was the classical solid state route. With this method you can obtain quite small particles in short time. But with this method we could not obtain nanopowders but it was used for comparison [9]. The precursors $\text{LiOH}\cdot\text{H}_2\text{O}$ (Fluka), $\text{Mg}(\text{AC})_2\cdot 4\text{H}_2\text{O}$ (Fluka), $\text{Ni}(\text{AC})_2\cdot 4\text{H}_2\text{O}$ (Aldrich) and $\text{Mn}(\text{AC})_2\cdot 4\text{H}_2\text{O}$ (Fluka) ($\text{AC} = \text{CH}_3\text{COO}$), were mixed with a blender in bain-marie at 80°C until the $\text{Mn}(\text{AC})_2\cdot 4\text{H}_2\text{O}$ was completely dissolved in its crystal water. The green solution obtained was put in an oven and calcinated. The calcination was done according to the below procedure to obtain best results:

- $10^\circ\text{C}/\text{min}$ until 500°C and rest for 5h, then $10^\circ\text{C}/\text{min}$ until 100°C and rest for 12h;
- $10^\circ\text{C}/\text{min}$ until 800°C and rest for 12h, then $1^\circ\text{C}/\text{min}$ until 100°C and rest for 12h. This procedure was repeated once again;
- $10^\circ\text{C}/\text{min}$ until 800°C and rest for 6h, then $0.1^\circ\text{C}/\text{min}$ until 100°C and rest for 12h.

The powders obtained were initially ground in a mortar before putting them in a planetary ball-mill with agate jars and balls (Fritsch Pulverisette 7) at the maximum speed-milling was done for 30 minutes.

2.1.2 Modified sol-gel

The sol-gel synthesis we adopted is a slight modification of the one used to create template TiO_2 [10]. The non-ionic triblock co-polymer Plurionic P123 surfactant (Aldrich), plus water, were stirred powerfully until its complete mixing. The chelating molecule *N*-butyl-ether ($\geq 99.5\%$, Aldrich) was added to prevent too fast condensation. The same precursors of the solid state synthesis were added to this solution. The solution was warmed at 80°C for one hour in bain-marie, and then stirred for 24h at room temperature before starting the calcination. Four steps were programmed without ball-milling in between, i.e.:

- $1^\circ\text{C}/\text{min}$ until 180°C and rest for 3h;
- $10^\circ\text{C}/\text{min}$ until 500°C and rest for 5h;
- $10^\circ\text{C}/\text{min}$ until 800°C and rest for 6h,
- $0.1^\circ\text{C}/\text{min}$ until 300°C and rest for 12h.

Finally the powders were ground in a mortar.

2.1.3 Xerogel

In this synthesis, a classical acid-catalyzed sol-gel method was applied. The same precursors of the solid state method plus EtOH anhydrous (J.T. Baker) were stirred in bain-marie at 60°C for 2h. After stirring, a solution of urea/EtOH/ H_2O (1:5:1 M) and 1 ml of HCl 0.3 M was added and stirred for 10 minutes. The solution was put in a crystallizer for one week at room temperature and for another week in a dry-oven at 80°C . The substance obtained was scratched, ground in a mortar and put in a oven for two thermal treatments:

- $10^\circ\text{C}/\text{min}$ until 500°C and rest for 5h;
- $10^\circ\text{C}/\text{min}$ until 800°C and rest for 6h, then $0.1^\circ\text{C}/\text{min}$ until 100°C and rest for 12h.

2.2 Coin-cell fabrication

After the synthesis, the powders need to be tested electrochemically. The synthesized active materials were mixed with polyvinylidene fluoride (Solvay), carbon black (99.9+%, $<1.0\mu\text{m}$; Alfa) and graphite ($\leq 0.1\text{mm}$; Fluka) with the following relative percentage ratios 80:10:3:7 and dissolved in 1-methyl-2-pyrrolydon (NMP, Merck-Schuchardt). This solution was treated in two ways. The first method was to ball-mill for 15 minutes and the pastes obtained were deposited in an aluminum strip with a $200\mu\text{m}$ thick doctor-blade and immediately dried at 80°C for a couple of hours. Then, after the aluminum strip was laminated until it reached a thickness of $70\mu\text{m}$, electrodes of $\varnothing 14\text{ mm}$ were cut. The other method was to mix the precursors for the paste by hand instead of using the ball-mill and then the same procedure as the first method was followed.

The electrochemical tests were done in CR2320 type coin-cells (Hohsen), assembled in this order: can, electrode, $\varnothing 21\text{ mm} \times 25\mu\text{m}$ separator; PP gas-ket, 4 drops of electrolyte (1 M LiPF_6 , EC/EMC/DMC Mitsubishi Chemical). $\varnothing 16\text{ mm}$ Lithium disk, $\varnothing 17\text{ mm}$ and 0.33 mm thick stainless steel spacer plate, $\varnothing 15\text{ mm}$ 1 ton pre-pressed wave spring and cap with insulating adhesive tape on top to prevent short circuits during the manipulation. The coin-cells were sealed in a He-filled glove-box ($\text{H}_2\text{O} < 5\text{ ppm}$). During the charge-discharge tests, the coin cells were kept under pressure with Hoffman clamps to ensure good electrical contacts.

2.3 Characterization techniques

To characterize the structure of our synthesis products, XRD patterns, SEM and TEM pictures were used which provide us some useful information about the dimension of the powders. Finally, coin-cell tests were performed.

2.3.1 Crystallographic structure

The structure of the powders were investigated with X-ray diffraction (XRD, Bruker AXS D8 advance, Cu-K_α), no filter at 45 kV and 40 mA , with a sample rotating frequency of 60 laps per minute. The typical scan type for a spinel structure was $10^\circ < 2\Theta > 70^\circ$. The powders were put in a PMMA disk with a copper-tin ring on the surface, and then the apparatus was put in the sample holder. Every diffraction pattern was taken without the $K_{\alpha 2}$ -Cu stripping [11]. The size of the crystals was estimated (but not calculated) by applying the Sherrer formula: it links the FWHM of a peak with the mean

crystallite size by:

$$T_{hkl} = \frac{0.89 \cdot \lambda}{\sqrt{L^2 - l^2} \cdot \cos(\Theta)} \quad (2.1)$$

where T is the size of the diffractive domain in the hkl direction ($\text{\AA}$); λ is the x-ray wavelength ($1.548 \text{ \AA}$ for Cu K_α); $L = \text{FWHM} \cdot 2\pi/180$ is the enlargement of a peak due to the diffraction domain; $l = \text{FWHM}' \cdot 2\pi/180$ is the contribution of the apparatus to the peak and Θ is the diffraction angle. For the Sherrer analysis we collected the FWHM of the most intense peak of the scan (in units of 2Θ).

2.3.2 Crystallinity and morphology

The crystallinity was studied with a transmission electron microscope (TEM) Philips CM30T with a LaB₆ filament as the source of electrons at 300 kV. Samples were mounted on Quantifoil® microgrid carbon polymer supported on a copper grid by placing ground sample on the grid. EDX elemental analysis was performed using a LINK EDX system. The copper signal in the EDX spectra is due to background radiation hitting the copper sample holder grid.

The particle size and morphology was examined with a SEM (JEOL JSM-5800LV) with different kV (10 or 15 kV) and different magnifications. The powders weren't gold-coated before the analysis.

2.3.3 Battery testing

The battery testing was performed with a Maccor S4000 tester at room temperature using different procedures to charge and discharge the batteries (see later).

Lithium-ion batteries were tested to study the voltage profile, the cycling behavior the cycle life and the ageing of the synthesized cathode material. The tests were done by galvanostatic¹ cycling between a potential range of 3.5-4.9 V. A constant charge-discharge current was applied to the battery, in terms of rate of capacity (C-rate). For example, a 1C-rate means that in one hour the battery is completely charged or discharged; with a 0.1C-rate (or C/10), we need 10 hours to charge or discharge a battery. The sum of a charge process plus a discharge one is called a *cycle*. The cycling of the cell is repeated for a number of times previously decided. Battery testing can

¹*galvanostatic* means that the test were done at constant current rate, and so voltage variations were studied

also be performed with different C-rates, to see the behavior of the cell in different stress conditions.

To apply a C-rate it is necessary to define a capacity “a priori” of the electrode. In our case we define a *theoretical capacity* in this way:

$$C = \frac{F \cdot X}{mw \cdot 3.6} \quad [C] = mAh/g \quad (2.2)$$

where F is the Faraday constant ($F = 96485C/mol$); X is the molar amount of lithium inserted in the spinel (we assume $X = 1$ as $LiNi_{0.5}Mn_{1.5}O_4$ can store 1 Li per mole of spinel), mw is the molecular weight of the active material and 3.6 is the conversion factor from A·s to mA·h.

When a charge-discharge process is performed, it is necessary divide the obtained capacity by the amount of active material of the electrode, in order to have comparable data. The quantity of active material is calculated in this way: we punched the weight of an aluminum disk of the same thickness and diameter of the electrode, and we subtracted its weight from the weight of the electrode. The number we obtained was multiplied for the percentage of active material used to create the paste (usually $80\% \Rightarrow \text{weight} \times 0.8$). The result is approximately the weight amount of active material inside the electrode ($w_{act-mat}$).

Chapter 3

Results

In this chapter we present the results of synthesis of $\text{LiMg}_\delta\text{Ni}_{0.5-\delta}\text{Mn}_{1.5}\text{O}_4$ cathode materials and the following coin-cell tests. The sections are divided for the δ -amount of magnesium in the stoichiometry of the cathode material ($\delta = 0.1, 0.07, 0.05$). SEM, TEM pictures are presented, with XRD patterns. Charge-discharge curves one C-rate are shown followed by only the discharge curves for the multiple C-rate graphs. Voltage vs capacity discharge plots are also shown: the data regarding the fifth cycle are collected for this, when the battery has stabilized.

3.1 $\text{LiMg}_{0.1}\text{Ni}_{0.4}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.1$

3.1.1 Solid State Synthesis

The cathode material was synthesized both with the solid state and modified sol-gel method. In every case the powders were dark grey with some orange part (sign of the presence of Li_2MnO_3) after the last oven process.

SEM and TEM analysis

As one can notice in fig 3.1, the particles are clustered. Nevertheless, the sample has a huge surface area - it looks like porous and rough.

The TEM pictures confirm that the sample is not nano-size and show that it isn't perfectly crystalline: in fig 3.2 and 3.3 you can see an amorphous shell present around the polycrystalline particle. From this point of view, it can cause a non perfect Li-insertion in the cathode. The lithium ions cannot find a suitable pathway to enter the domain, so they spend more time to arrive in the crystalline part of the material.

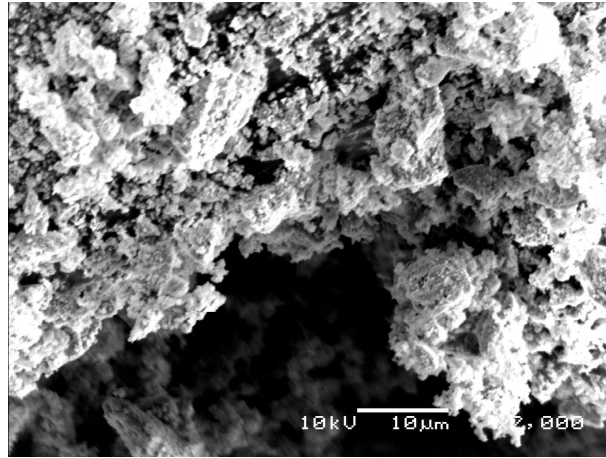


Figure 3.1: SEM picture (magn. 2000) of $\delta = 0.1$ sample.

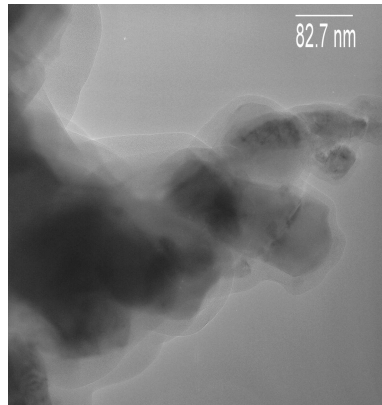


Figure 3.2: TEM picture of $\delta = 0.1$ sample.

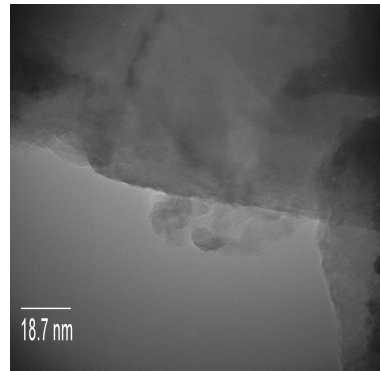


Figure 3.3: TEM picture of $\delta = 0.1$ sample - detail.

X-ray diffraction

Figure 3.4 shows the XRD pattern of $\text{LiMg}_{0.1}\text{Ni}_{0.4}\text{Mn}_{1.5}\text{O}_4$. The spinel structure is formed but eventual impurities like $\text{Li}_x\text{Ni}_{1-x}\text{O}$ cannot be seen: such lithium nickel oxide presents two very little peaks at 37.5 and 43.5 values of 2θ angle and maybe are completely covered by the baseline [9] or by the spinel peak as Li_2MnO_4 .

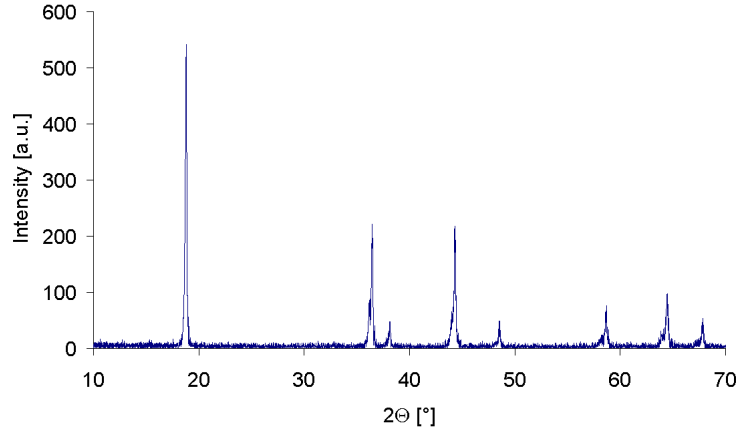


Figure 3.4: XRD pattern of $\delta = 0.1$ cathode material.

Electrochemical measurements

For the $\delta = 0.1$ sample created with the solid state method, a C/10 and C/4 charge-discharge rate were adopted. Its theoretical capacity is 149.5 mAh/g. It can be seen that with the mortar procedure to obtain the paste, the results are not good; the overall capacity is fading. This is probably due to a non perfect mixing of the precursors (see 3.5 and 3.6).

Anyway the theoretical capacity of $\text{LiMg}_{0.1}\text{Ni}_{0.4}\text{Mn}_{1.5}\text{O}_4$ is not reached even with the ball-milling method for the paste (see 3.7 and 3.8). This is due moreover to the non good purity of the synthesized powders and to the amorphous layer covering the crystals. Moreover it's remarkable the fact that $C_{C/4} > C_{C/10}$; it can be an indication that the mixing could be wrong.

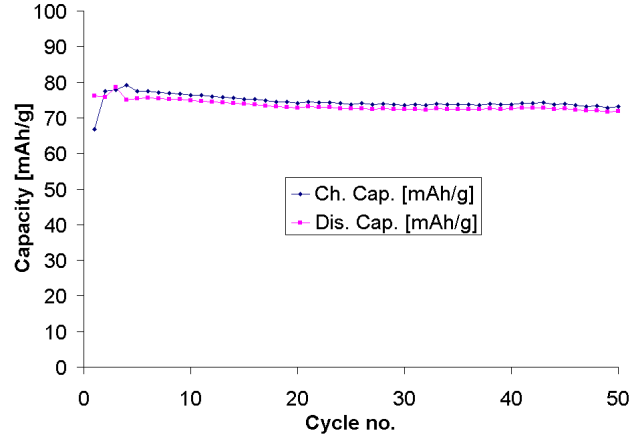


Figure 3.5: C/10 Charge-discharge plot of $\delta = 0.1$ sample. The paste for the electrode was created by hand with the mortar.

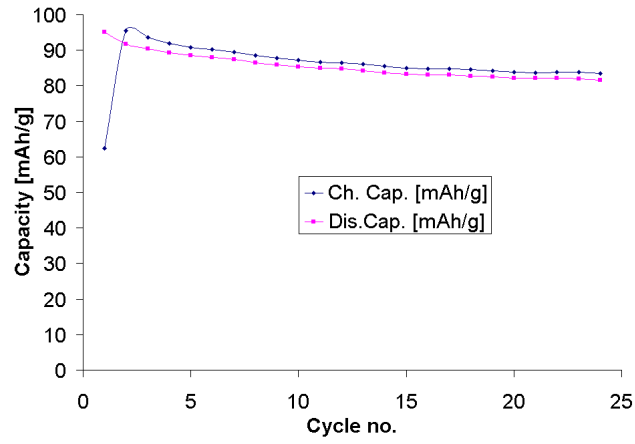


Figure 3.6: C/4 Charge-discharge plot of $\delta = 0.1$ sample. The paste for the electrode was created with the mortar.

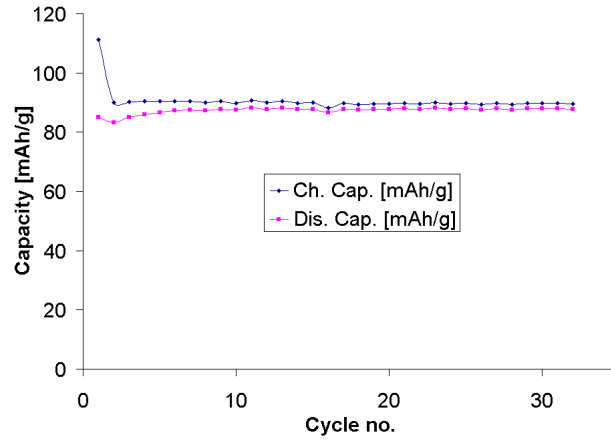


Figure 3.7: C/10 Charge-discharge plot of $\delta = 0.1$ sample. The paste for the electrode was created with the ball-miller.

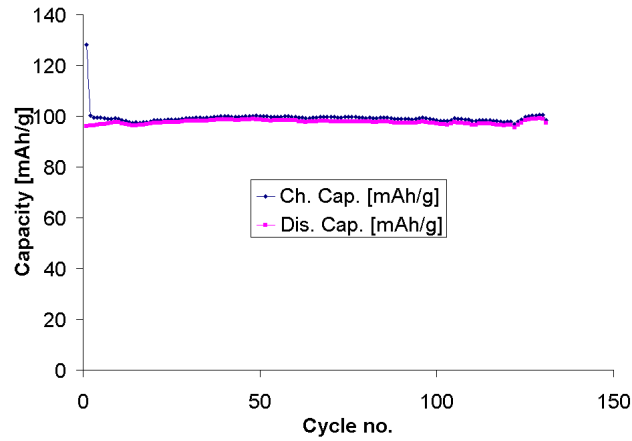


Figure 3.8: C/4 Charge-discharge plot of $\delta = 0.1$ sample. The paste for the electrode was created with the ball-miller.

In the next graphs two (fig. 3.9 and 3.10), voltage-capacity curves taken at C-10 during the fifth discharge cycle with two different paste method are shown. Two plateau are present: one around 4.7-4.8V, regarding the oxidation of nickel ($\text{Ni}_{0.5-\delta}^{2+} \rightarrow \text{Ni}_{0.5-\delta-x}^{2+} + \text{Ni}_x^{4+}$) and the other around 3.9-4.1V, probably a sign of the presence of the spinel LiMn_2O_4 that is responsible of the reversible reaction is $\text{Mn}_{0.5}^{3+} + \text{Mn}_{0.5}^{4+} \rightarrow \text{Mn}_{0.5-\alpha}^{+3} + \text{Mn}_{0.5+\alpha}^{4+}$ [9, 12]

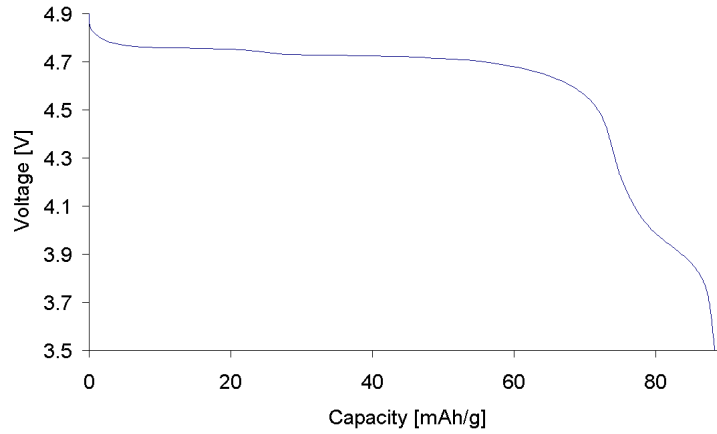


Figure 3.9: C/10 voltage vs capacity plot of $\delta = 0.1$ sample.
(paste = ball-milling method).

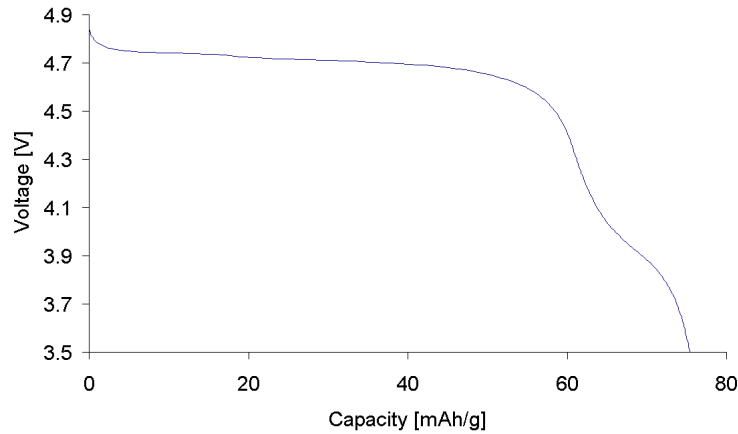


Figure 3.10: C/10 voltage vs capacity plot of $\delta = 0.1$ sample.
(paste = mortar method).

In every experiment, the mortar procedure to obtain the paste for the cathode gave poor results, so it was no longer used, instead we have used the ball-milling process. Therefore, the next plots of electrochemical results will relate only to the latter procedure.

3.1.2 Sol-gel Synthesis

Two similar successive thermal treatments were performed for this synthesis, in order to let the parasitic phase LiMn_2O_4 disappear. the samples were brought to 800°C and slowly cooled to 300°C ($0.1^\circ\text{C}/\text{min}$). A 30 min. ball-milling procedure was performed after each thermal treatment.

TEM analysis

SEM has not been performed, instead TEM measurements were done on this samples, and it is clear from fig 3.11, 3.12, 3.13, 3.14, that the size distribution is very large: crystalline particles are present from around 30 nm up to more than 200 nm, and always an amorphous layer around the crystals is present. You can observe the cubic spinel structure from the TEM pictures.

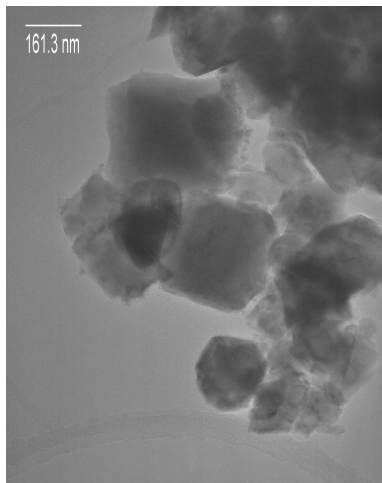


Figure 3.11: TEM picture of $\delta = 0.1$ with sol-gel method and first thermal treatment.

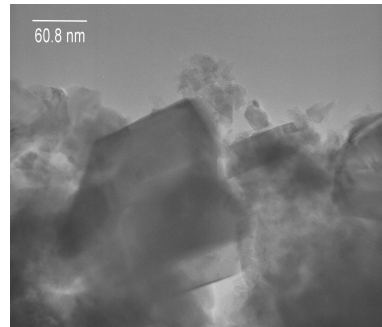


Figure 3.12: TEM picture of $\delta = 0.1$ with sol-gel method and first thermal treatment - detail.

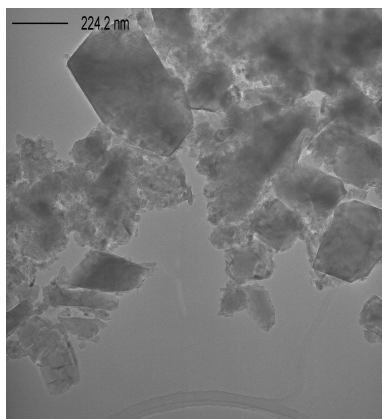


Figure 3.13: TEM picture of $\delta = 0.1$ with sol-gel method and second thermal treatment.

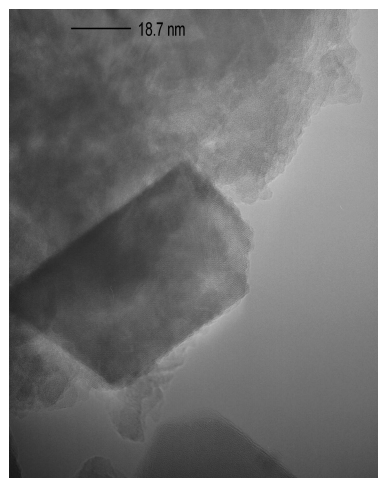


Figure 3.14: TEM picture of $\delta = 0.1$ with sol-gel method and second thermal treatment - detail.

X-ray diffraction

In the picture 3.15 the XRD scans with and without second thermal treatment are shown. From the patterns, no suitable changes can be shown, due to the two successive thermal treatments.

Electrochemical measurements

The figure 3.16 shows the plots V vs C of the $\text{LiMg}_{0.1}\text{Ni}_{0.4}\text{Mn}_{1.5}\text{O}_4$ with and without the second thermal treatment. As you can notice, the second thermal treatment lower the capacity. Nevertheless you can see that the shoulder related to the LiMn_2O_4 phase it's smaller in the second thermal treatment sample. Indeed, the successive thermal treatment leads to a better pureness of the sample, decreasing the LiMn_2O_4 phase. Anyway, the successive thermal treatment leads to an increase of the particle size too, and bigger is the particle, less capacity it can exchange in a charge-discharge experiment.

The same results you can observe in fig. 3.17 where capacity versus cycle number is plotted. Here the data points are grouped per C-rate. In general, the lower the C-rate, the higher is the capacity because lithium has enough time to fill all the vacancies of the host spinel. Typically the capacity of the material with only the first thermal treatment is higher than the other one.

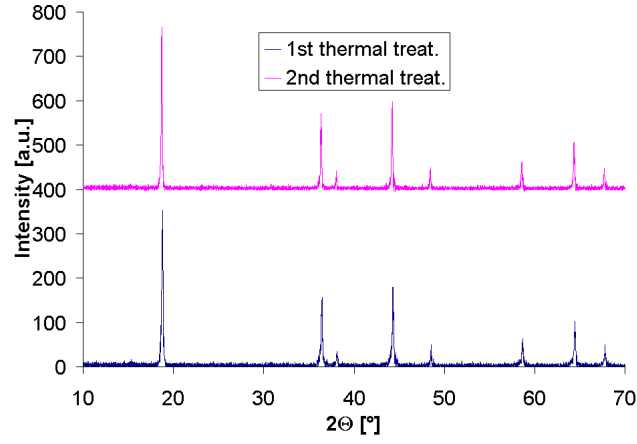


Figure 3.15: XRD patterns after the first (blue) and after the second (pink) thermal treatment.

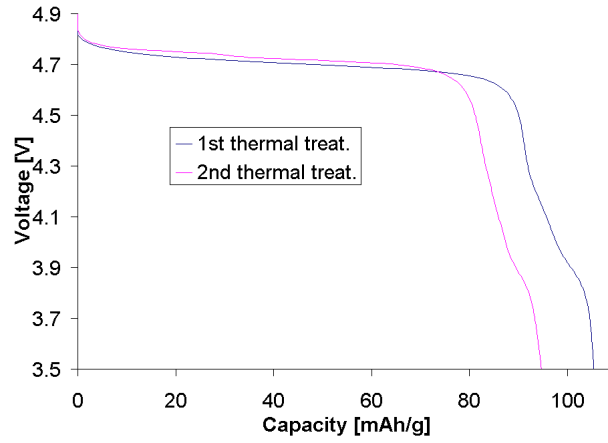


Figure 3.16: Voltage vs capacity of two coin-cells with the $\delta = 0.1$ active material with first and second thermal treatment.

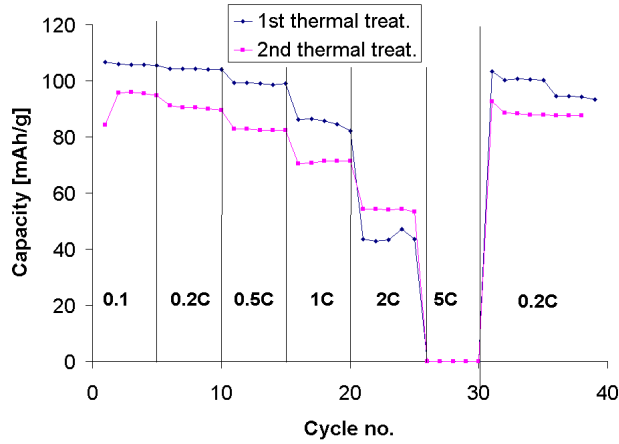


Figure 3.17: Capacity related to the number of charge-discharge cycles of two coin-cells with the $\delta = 0.1$ active material with first and second thermal treatment.

3.2 $\text{LiMg}_{0.07}\text{Ni}_{0.43}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.07$

3.2.1 Solid State Synthesis

The spinel $\text{LiMg}_{0.07}\text{Ni}_{0.43}\text{Mn}_{1.5}\text{O}_4$ was synthesized only with the solid state method because this material gave the worst electrochemical results. Also in this case an orange impurity was present, indicating the presence of Li_2MnO_3 .

SEM and TEM analysis

The SEM pictures show high porosity of the $\delta = 0.07$ material, but the powders are still big, as you can see in fig. 3.18, 3.19. In fact, even in this case the size distribution is large and hard to analyze with TEM. Nevertheless, TEM analysis in fig 3.20 and 3.21, in which big cubic crystals (even if smaller than the $\delta = 0.1$ pictures showed) are covered by an amorphous material that may prevent a high capacity. Anyway the domain size are from 20nm until 200 nm, confirming the large size distribution.

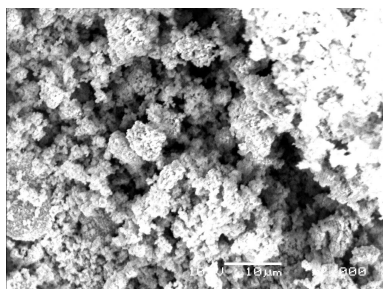


Figure 3.18: SEM picture of $\delta = 0.07$ with solid state method, magnification 2000.

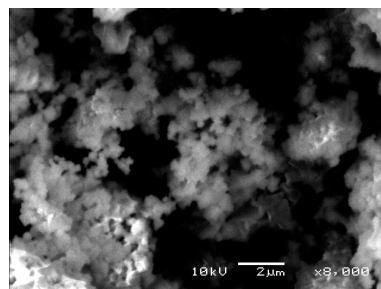


Figure 3.19: SEM picture of $\delta = 0.07$ with solid state method, magnification 8000.

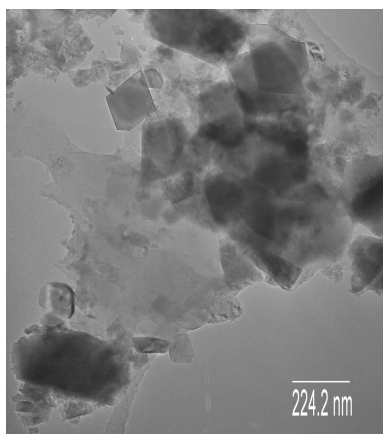


Figure 3.20: TEM picture of $\delta = 0.07$ with solid state method.

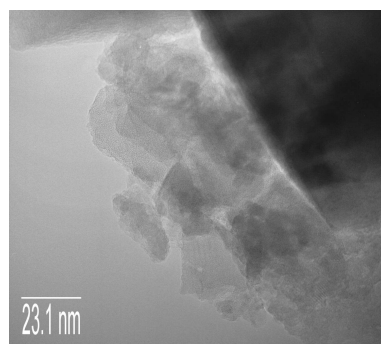


Figure 3.21: TEM picture of $\delta = 0.07$ with solid state method - detail.

X-ray diffraction

The XRD pattern shows that the spinel is formed, but the FWHM of the peaks suggests that the particles are quite small, in agreement with the previous TEM pictures.

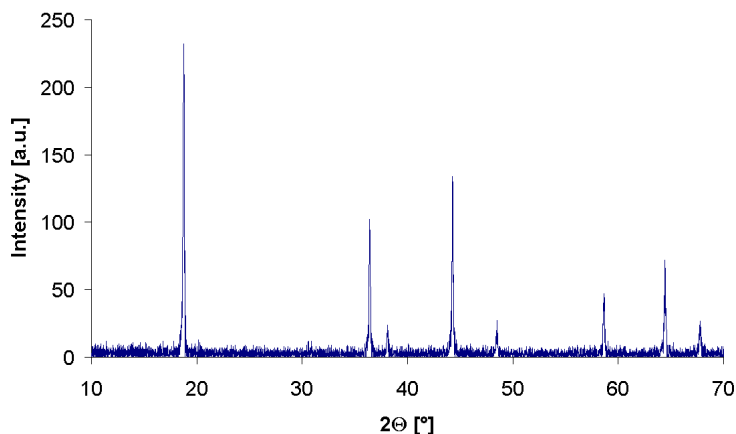


Figure 3.22: X-ray pattern of $\delta = 0.07$ stoichiometry.

Electrochemical measurements

A C/4 procedure of charge-discharge was performed and fig. 3.23 shows that the capacity of the charge and discharge are not constant, but they continue to decrease. It means that the coin-cell is not stable, due to a bad assembly of the coin-cell or to a not perfect synthesis of the material.

However, figure 3.24 suggests that the problem is not the material, but then maybe the coin-cell: it is remarkable that in the region of the oxidation of the manganese, the plateau is very small and it means that there is few LiMn_2O_4 present compared to the $\delta = 0.01$ material.

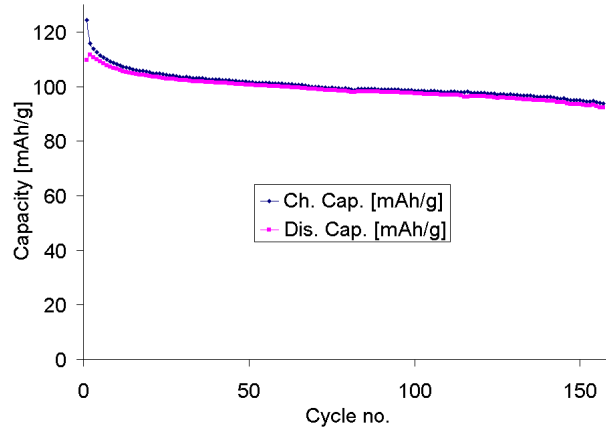


Figure 3.23: Charge-discharge curves with C/4 C-rate of $\delta = 0.07$ material.

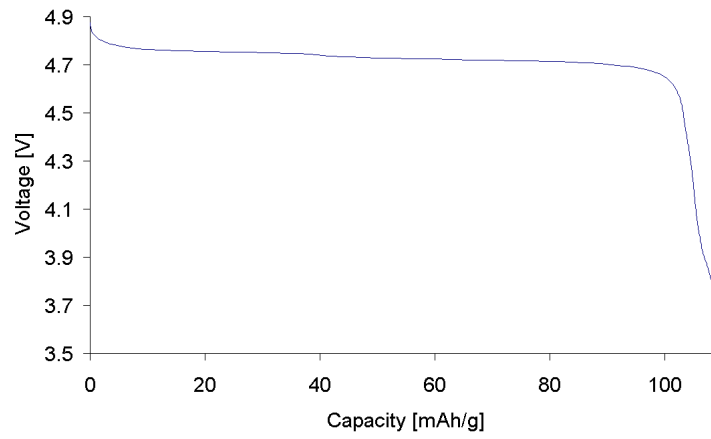


Figure 3.24: Voltage versus Capacity of $\delta = 0.07$ material.

3.3 $\text{LiMg}_{0.05}\text{Ni}_{0.45}\text{Mn}_{1.5}\text{O}_4$: $\delta = 0.05$

The spinel $\text{LiMg}_{0.05}\text{Ni}_{0.45}\text{Mn}_{1.5}\text{O}_4$ was synthesized with all the techniques described before: the solid state, the sol-gel and the xerogel. It gave the best results in terms of dimension of the powders and of capacity of the batteries.

3.3.1 Solid State Synthesis

SEM and TEM analysis

The SEM picture of the $\text{LiMg}_{0.05}\text{Ni}_{0.45}\text{Mn}_{1.5}\text{O}_4$ material show high porosity but big powders (fig. 3.25 and 3.26), as confirmed by the TEM analysis. For example, in the figure 3.27 very big and crystalline grains are present, while in the fig. 3.28 you can see only a border of a bigger crystal. Hence, the primary particle is almost the domain (grain). Fig. 3.27 shows even a quite homogeneous size distribution of the crystals.

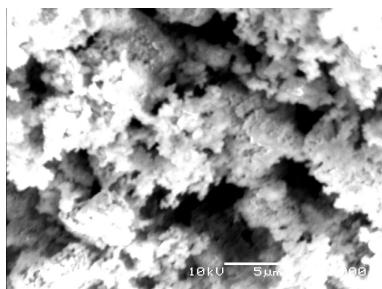


Figure 3.25: SEM picture of $\delta = 0.05$ with solid state method, magnification 4000.

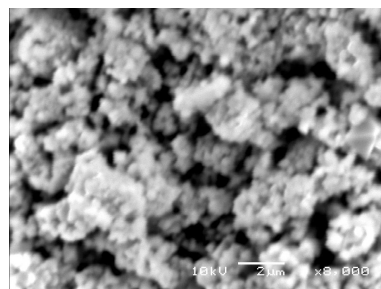


Figure 3.26: SEM picture of $\delta = 0.05$ with solid state method, magnification 8000.

X-ray diffraction

The very narrow peaks (small FWHM) in the XRD pattern shown in fig 3.29 supports the thesis about the size of particles.

Electrochemical measurements

The strange behavior of the charge plot in fig 3.30 can be related to a non perfect assembly of the coin-cell. It Sometimes happens that you touch the negative and positive parts of the battery at the same time with the metal tools. So a short circuit is created and the battery loose potential. In both of

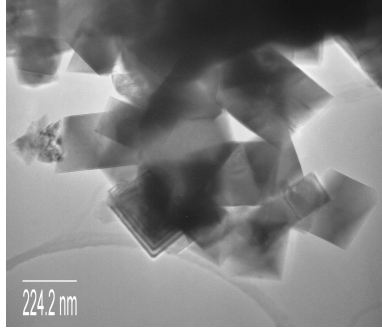


Figure 3.27: TEM picture of $\delta = 0.05$ with solid state method.

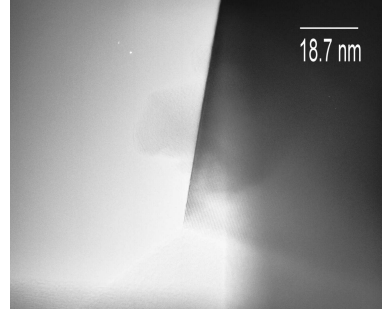


Figure 3.28: TEM picture of $\delta = 0.05$ (solid state) - detail.

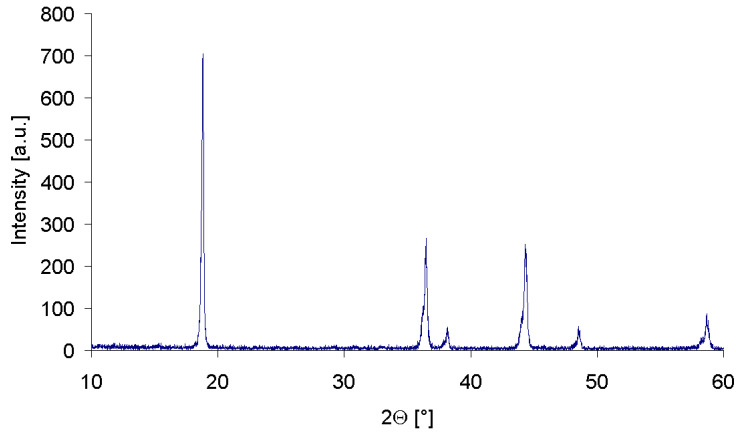


Figure 3.29: X-ray pattern of $\delta = 0.05$ material.

the graphs (3.30 and 3.31), the curves are not flat, but they decrease towards lower values of capacity. It means that the battery is not stable. The fig. 3.32 suggest that there is a problem of the fabrication of the battery: in that graph a very flat potential around 4.8V is visible (Ni oxidation) with only a minor part at 3.9-4.1V, hence, is little: there's little LiMn_2O_4 in the system.

3.3.2 Sol-gel Synthesis

Much better results were obtained with the sol-gel $\delta = 0.05$ powders, regarding both the particle dimensions and electrochemical results. Two thermal treatments were performed on the powders, (both of them at 800°C with a cooling rate of 0.1°C/min to 300°C finished by a 30 min of ball-milling after

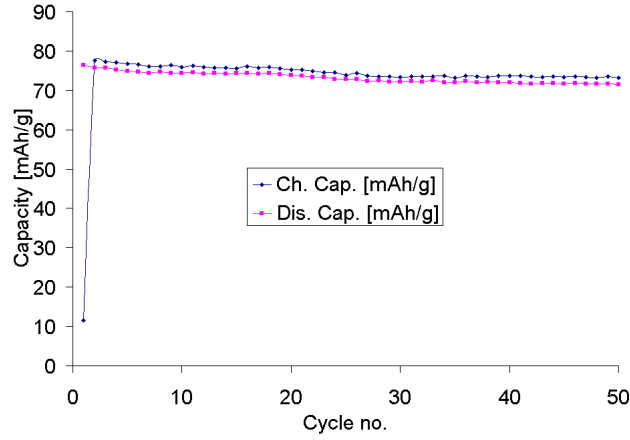


Figure 3.30: Charge-discharge plot of $\delta = 0.05$ material at $C/10$.

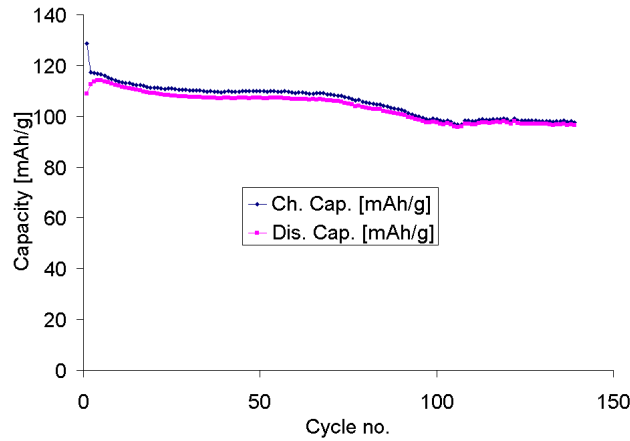


Figure 3.31: Charge-discharge plot of $\delta = 0.05$ material at $C/4$.

each treatment) to get rid of the amorphous phase always present in our samples. SEM pictures are related to the first thermal treatment, whereas TEM ones are taken of both thermal treated samples.

SEM and TEM analysis

The SEM pictures (3.33, 3.34) show a very high porosity of the material, even at high magnification where you can see small grains. The TEM analysis of

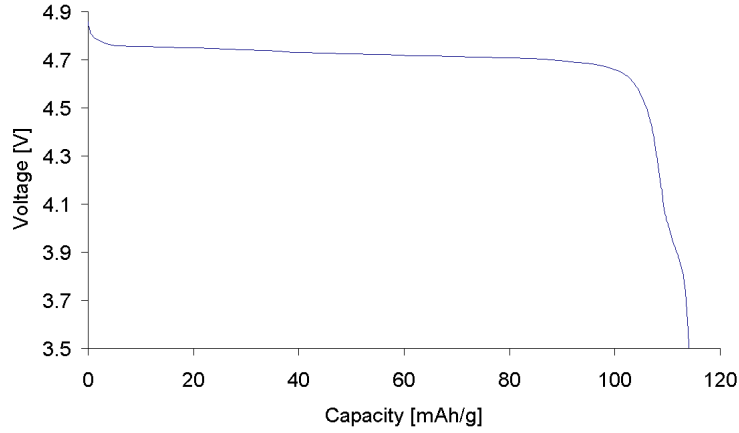


Figure 3.32: V vs. C plot of $\delta = 0.05$ solid state method at C/4.

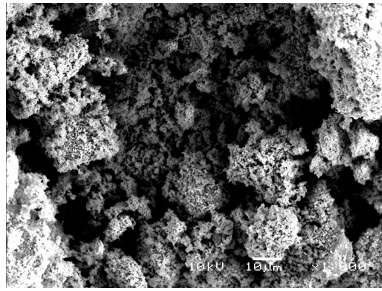


Figure 3.33: SEM picture of $\delta = 0.05$ with sol-gel method, magn. 1000.

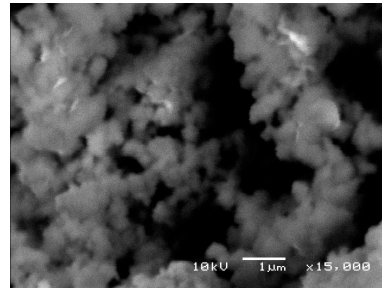


Figure 3.34: SEM picture of $\delta = 0.05$ with sol-gel method, magni. 15000.

the sample measured immediately after the first thermal treatment, exhibits a large size distribution: the powders are in the nanoscale, and do show the presence of an amorphous phase (3.35 3.36).

It can be seen from pictures 3.37 and 3.38 that the second thermal treatment enlarges the particle size without complete removal of the amorphous layer around the particles. The size of the crystals are from 10 nm until 200 nm.

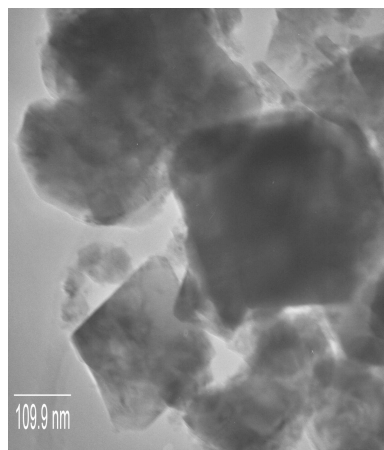


Figure 3.35: TEM picture of $\delta = 0.05$ with sol-gel method, 1st therm. treat.

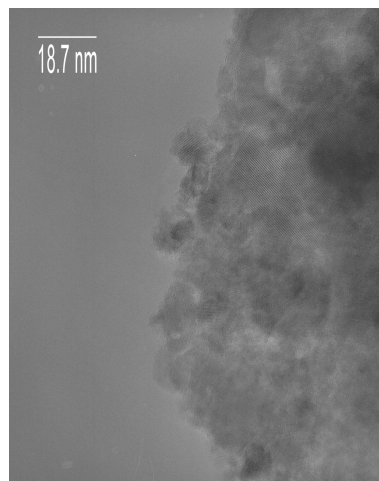


Figure 3.36: Detail of the previous picture.

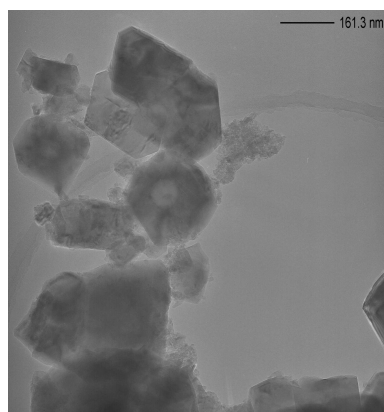


Figure 3.37: TEM picture of $\delta = 0.05$ with sol-gel method, 2nd therm. treat.

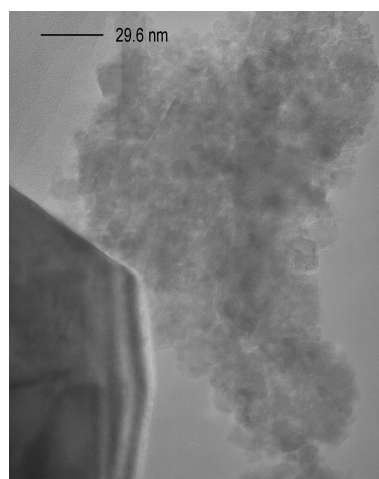


Figure 3.38: Detail of the previous picture.

X-ray diffraction

The XRD patterns confirm again what has been discussed before regarding the size of the particles: the peaks are very narrow compared to the XRD of the other methods, but the XRD of the second thermal treatment exhibits the largest FWHM.

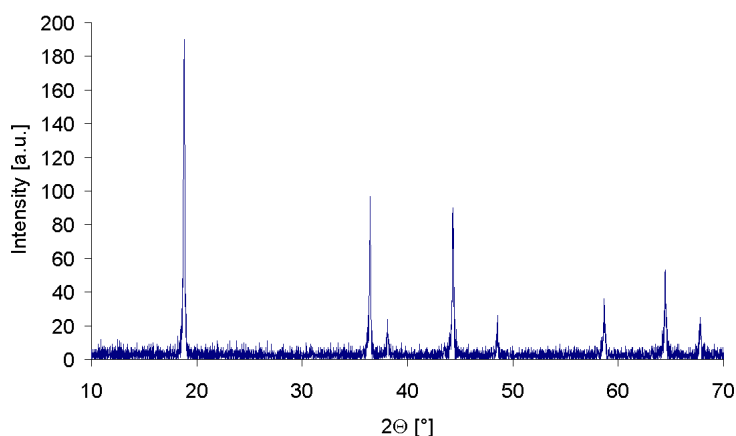


Figure 3.39: X-ray pattern of $\delta = 0.05$ material sol-gel - 1st therm. treat.

Electrochemical measurements

In fig. 3.41 various discharge rates plot are shown in which we compare the sample with the two thermal treatments. There is a big difference between the two curves in terms of capacity. The material with only one treatment has a much higher capacity at every C-rate than the sample treated twice. The reason for this is most likely the increase of the size of the particles.

Fig. 3.42 is related to the Voltage-Capacity plot. Here you can see how much LiMn_2O_4 disappeared with the second treatment (the shoulder related to the Mn^{3+} oxidation, is smaller) and, on the other hand, how much the capacity is decreased.

3.3.3 Xerogel Synthesis

The last route we used to create cathode materials was the xerogel synthesis. In this case a dark brown very hard gel was the result of the dry-oven oven procedure at 80°C. After scratching this gel SEM, TEM and XRD analysis

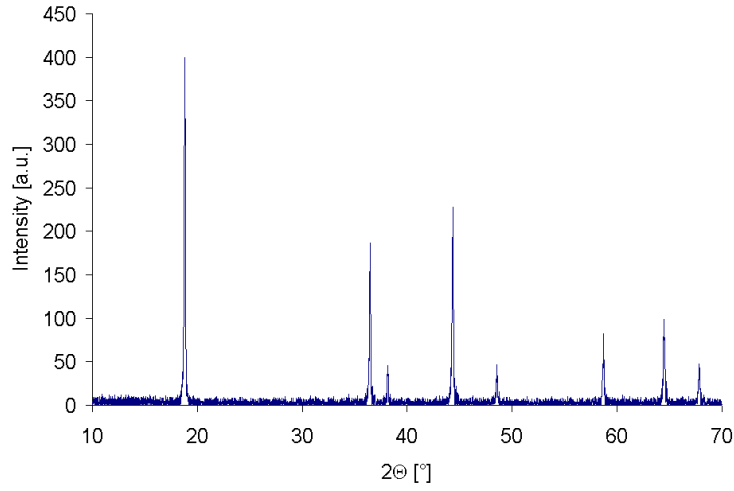


Figure 3.40: X-ray pattern of $\delta = 0.05$ material sol-gel - 2nd therm. treat.

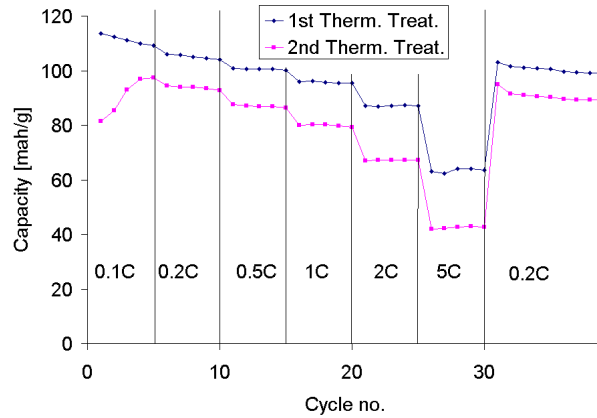


Figure 3.41: Discharge curves of $\delta = 0.05$ material at various discharge rates.

were performed. Other TEM images and XRD plots were taken after every successive thermal treatment, to study the growth of the crystals.

SEM and TEM analysis

Only SEM pictures of no thermal treatment $\delta = 0.05$ xerogel material were taken (3.43, 3.44). The powders are very big and almost not porous. The

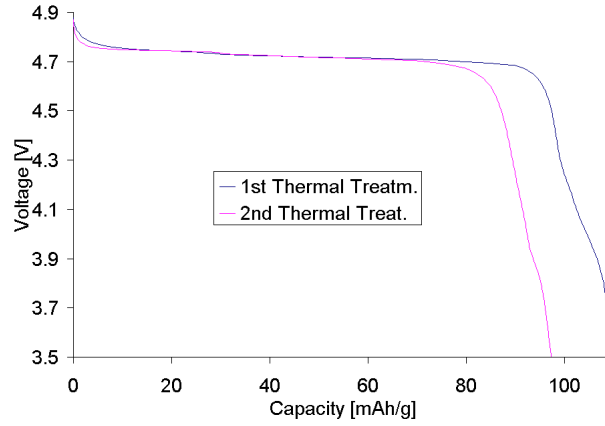


Figure 3.42: V vs C curves of $\delta = 0.05$; 1st and 2nd therm. treat.

TEM pictures of these powders (3.45, 3.46) show enormous crystals with a big part of amorphous material around it. After the first treatment at 500°C, the situation changes: in fig. 3.47 and 3.48 little particles are presents. Fig. 3.49 and 3.50 show TEM pictures of the xerogel material after the second thermal treatment. Bigger crystals are present, but also little ones still remain.

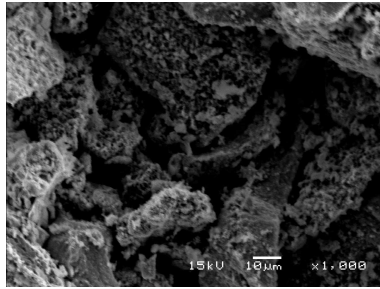


Figure 3.43: SEM picture of $\delta = 0.05$, xerogel method without any thermal treatment, magn. 1000.

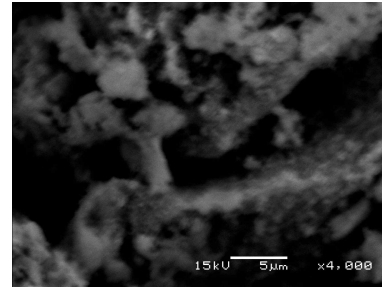


Figure 3.44: SEM picture of $\delta = 0.05$, xerogel method without any thermal treatment, magn. 4000.

X-ray diffraction

XRD patterns of all the three samples related to the xerogel process are shown in fig 3.51. The sample without any calcination isn't a spinel: the

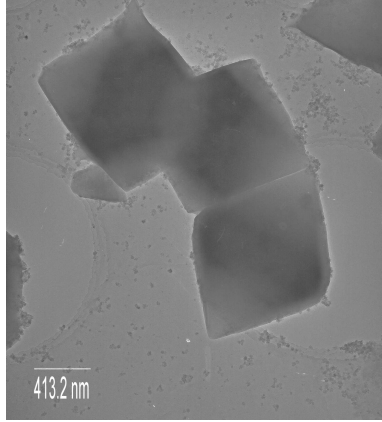


Figure 3.45: TEM picture of $\delta = 0.05$, xerogel method without any thermal treatment.

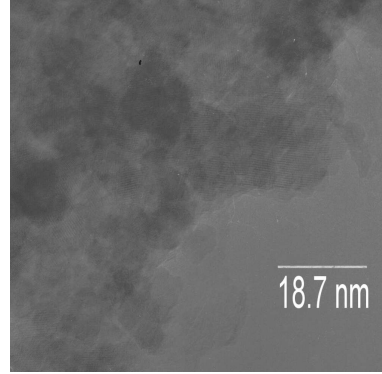


Figure 3.46: TEM picture of $\delta = 0.05$, xerogel method without any thermal treatment - detail.

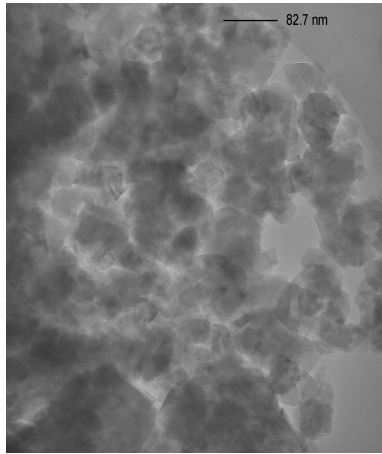


Figure 3.47: TEM picture of $\delta = 0.05$, xerogel method with thermal treatment at 500°C.

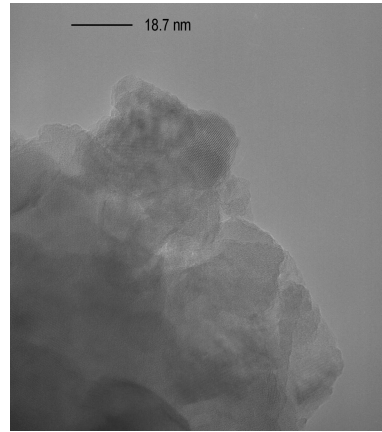


Figure 3.48: TEM picture of $\delta = 0.05$, xerogel method with thermal treatment at 500°C - detail.

80°C temperature of the dry-oven is not enough to form such a structure. At 500°C a spinel is visible, having particles around 30-50 nm. When the sample is heated to 800°C, the spinel is crystallized more and more, reflected by a narrower FWHM.

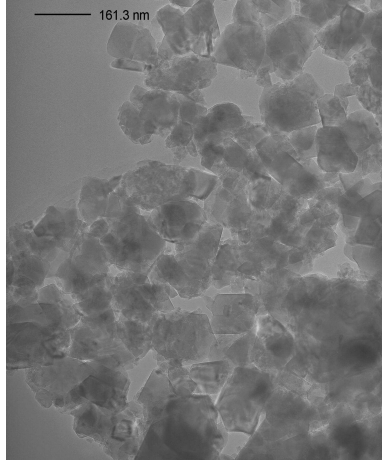


Figure 3.49: TEM picture of $\delta = 0.05$, xerogel method with thermal treatment at 800°C.

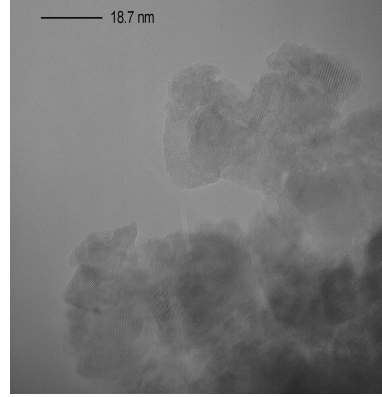


Figure 3.50: TEM picture of $\delta = 0.05$, xerogel method with thermal treatment at 800°C - detail.

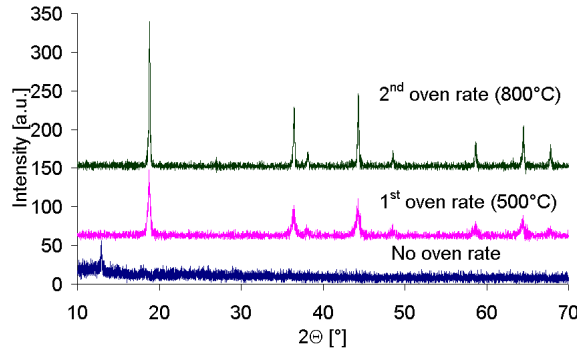


Figure 3.51: X-ray pattern of $\delta = 0.05$ material with xerogel method and the three different processes.

Electrical measurements

In fig. 3.52 the discharge plot is shown. The material has a quite high capacity even at high C-rates. Besides, the capacity at 0.2C is the same in both of its intervals: it means that the battery can support stress due to the continuous charge-discharge even at different C-rates. The next graph (3.53) exhibits a very flat plateau on the Ni oxidation region although the plateau

due to the oxidation on Mn is significant.

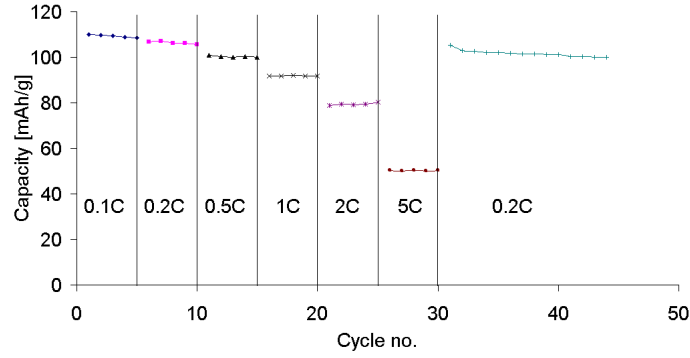


Figure 3.52: Plot of discharge curves at different C-rates of $\delta = 0.05$ material with xerogel method after the second rate oven at 800°C .

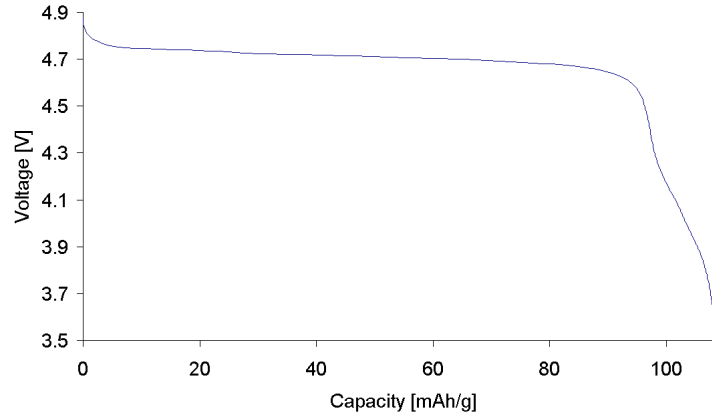


Figure 3.53: V vs C plot of $\delta = 0.05$ material after the second rate oven at 800°C .

Chapter 4

Summary and Conclusions

In order to synthesize cathode materials with better performance, this research was focused on the fabrication of nanopowders. The material selected for that purpose was the high-voltage spinel $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$. The synthesis and the characterization of the high voltage spinel $\text{LiMg}_\delta\text{Ni}_{0.5-\delta}\text{Mn}_{1.5}\text{O}_4$ ($\delta=0.05, 0.07, 0.1$) was performed. In this work we try to find relationships between the synthesis process, the powder morphology and the electrochemical performance. In order to find different powders morphologies, three different synthesis techniques were employed, i.e. solid state route, sol-gel and xerogel methods.

The synthesis of the powders lead to non pure materials. In fact all the powders made contained an orange impurity which is probably Li_2MnO_3 . Besides, $\text{Li}_x\text{Ni}_{1-x}$ and LiMn_2O_4 may be present. All the three impurities gave X-ray diffraction peaks at the same angles. The presence of LiMn_2O_4 is confirmed by the voltage vs capacity plot.

The synthesis via the sol-gel and xerogel routes typically reveals nanopowders with a very large size distribution. Nevertheless, homogeneous size distribution from 20 to 200 nm can be expected by proper thermal treatments. The synthesis via solid state reactions has shown a narrow size distribution, where the primary particles are often single crystals which have sizes between 200-300 nm. These techniques allow us to synthesize a range of powders from 20 to 200 nm of single crystals.

If we put the Full Width at Half Maximum (FWHM) of the most intense peak of each XRD pattern for $\text{LiMg}_{0.05}\text{Ni}_{0.45}\text{Mn}_{1.5}\text{O}_4$ versus the inverse of the mean of the particles size (fig 4.1), we clearly can see a trend without being accurate, this trend is roughly similar to the trend one can expect on the basis of the Scherrer equation.

It is further stressed that the errors in the particle size is huge as they are estimated from the TEM pictures, whereas the error in the FWHM is

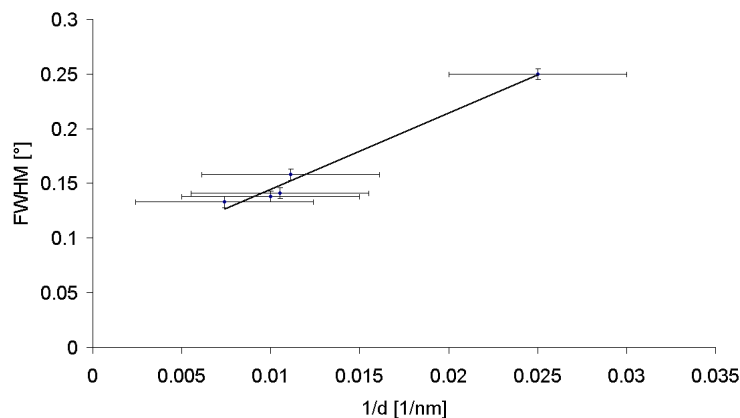


Figure 4.1: FWHM values versus the inverse of the particle size.

relatively small as they have been calculated by the XRD fitting software.

The battery tests shown good cyclability, particularly for the $\delta = 0.05$ material. The large powders made with the solid state method reveal a pretty good cyclability, but the best results in terms of capacity, rate capability and stability were obtained with the sol-gel synthesis. With optimized material these results could even be improved. In addition, the xerogel route is believed to be very promising. Particularly, with the use of urea we may expect significant improvements, as this molecule produces a very exothermic reaction when it is fired. Further with this method, we've shown that, in principle, to form a spinel it is not necessary to apply high temperatures. however, optimal result were not obtained and additional research is still necessary. Actually, the xerogel route can be seen as another type of sol gel, leading to the same electrochemical results.

The $\delta = 0.07$ material didn't show acceptable electrochemical behavior. The material was not pure, but this may be improved by the novel xerogel method or even the sol-gel method. Nevertheless, for scientific reason, it is important to have at least a pure composition to compare with the The $\delta = 0.05$ material.

The highest Mg doped spinel $\delta = 0.1$ showed the poorest results, for the powders made via the solid state method; the discharge curves as a function of the cycle number show significant fading and the initial capacity is not all close to the theoretical one. The large amount of LiMg_2O_4 present as impurity, may cause this fading, but a poor closing of some coin-cells may not be excluded. We obtained much better results with the sol-gel synthesis, in terms of capacity and cyclability.

In summary, with the three synthesis techniques, we were able to synthesize almost nanodispersed particles that are practically single crystals, having sizes between 20 to 200 nm depending on the synthesis method. To date, the sol-gel technique showed the best results. However the xerogel method employing urea is a promising alternative. It is expected that with the last method, capacities close to the theoretical capacity can be made with particles down to 20 nm, necessary for high power applications.

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Doei Doeiiii.

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